AI ROCKS THE CRADLE

From egg and sperm selection to embryo grading, AI is enhancing IVF success rates and bringing joy to hundreds of childless couples. All it needs now are better datasets and new regulatory laws

By **SONALI ACHARJEE**
Illustration by **NILANJAN DAS / AI**



THE SPREAD OF ARTIFICIAL INTELLIGENCE ACROSS OUR PROFESSIONAL AND DOMESTIC LIVES PREPARES us for a future that must be in lockstep with the rhythms of algorithms. Now, its nimble, knowing fingers are touching the origins of life. Several months ago, a couple from Dehradun walked into the Birla Fertility Clinic in Gurugram. The woman had endured two molar pregnancies—a complication in which abnormal tissue and not a foetus grows in the uterus—each ending with a painful dilation and curettage (D&C), a procedure to clear the uterine lining. The trauma caused Asherman's syndrome, where scar tissues form inside the uterus, impeding pregnancy. Doctors recommended In-vitro Fertilisation (IVF), where a woman's egg is fertilised in a lab with sperm, and the resultant embryo transferred to her uterus to achieve pregnancy. But first the woman needed to be healed. Obligatory genetic tests were done, followed by a hysteroscopy to remove the scar tissues and sessions of PRP therapy, where platelet-rich plasma were drawn from her own blood for tissue repair. Slowly, her uterine lining strengthened and a genetically normal embryo was transferred. Months later, she held a baby girl in her arms. Each procedure in this happy IVF story was perfected through a powerful new ally: artificial intelligence.

As fertility clinics across India strive to improve success rates, they are turning to algorithms to tilt the odds of conception in patients' favour. From analysing embryo quality with more precision than the human eye to predicting ovarian response and tailoring stimulation protocols (personalised fertility medication), AI is sharpening IVF procedures. "The efficiency of IVF treatment is continuously improving, and one of the most promising advancements is the integration of AI at various stages," says Dr Manjunath, deputy medical director, Birla Fertility and IVF, Bengaluru.

Once disparaged for creating 'artificial babies', IVF has become a lifeline for Indian couples facing difficulty in conceiving. The In-

